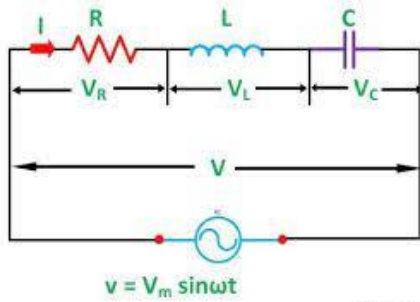
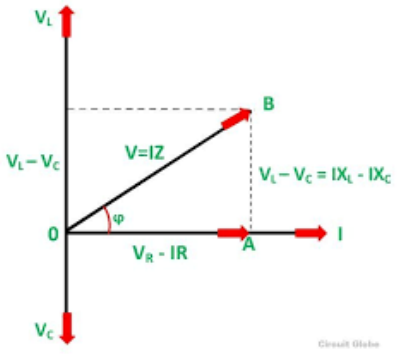


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021		
Course Code: EST130		
Course Name: BASICS OF ELECTRICAL & ELCTRONICS ENGINEERING		
PART I: BASIC ELECTRICAL ENGINEERING		
PART A		
		<i>Answer all questions. Each question carries 4 marks</i>
Q1		A coil of 180 turns is linked with a flux of 0.03 Wb when carrying a current of 10A. Calculate the inductance of the coil. If the current is uniformly reversed in 0.04 sec, calculate the emf induced in the coil.
Ans		$L = \frac{N\phi}{i} = \frac{180 \times 0.03}{10} = 0.54H$ When current is reverse, change in current $di = 20A$ $e = L \frac{di}{dt} = 0.54 \times \frac{20}{0.04} = 270V$
Q2		An alternating current is represented by $i(t) = 14.14 \sin(377t)$. Find (i)rms value (ii) frequency (iii)time period and (iv)instantaneous value of the current at $t = 3ms$.
Ans		$i(t) = 14.14 \sin(377t)$ comparing with the general representation of alternating current, $i(t) = I_m \sin(\omega t + \phi)$ $I_m = 14.14A$ $\omega = 377 \text{ rad/sec}$ (i) $RMS \text{ Value} = \frac{I_m}{\sqrt{2}} = \frac{14.14}{\sqrt{2}} = 10A$ (ii) $\omega = 2\pi f$ $frequency, f = \frac{\omega}{2\pi} = \frac{377}{2\pi} = 60Hz$ (iii) Time period, $T = 1/f = 0.0167 \text{ Sec}$ (iv) Instataneous value of current at $t = 3ms = 0.003 \text{ sec}$ $i(t) = 14.14 \sin(377t)$ $i(0.003) = 14.14 \sin(377 \times 0.003) = 12.794A$
Q3		Derive an expression for the energy stored in an inductor.

Ans	Energy, $E = \int P dt = \int V I dt$ For an inductor, $V = L \frac{di}{dt}$ Thus, $E = \int L \frac{di}{dt} \times i dt = \int L i di$ $E = L \frac{i^2}{2}$ Thus, Energy stored in an inductor $E = \frac{1}{2} L i^2$
Q4	Derive the expression for the current in an ac series RLC circuit.
Ans	  Applying parallelogram law, $V = \sqrt{V_R^2 + (V_L - V_C)^2}$ $V = \sqrt{(IR)^2 + (IX_L - IX_C)^2} = I \sqrt{R^2 + (X_L - X_C)^2} = IZ$ $I = \frac{V}{Z}$ $i(t) = I_m \sin(\omega t - \phi)$
Q5	A resistance of 10Ω , an inductance of 0.3 H , and a capacitance of $100\mu\text{F}$ are connected in series across 230V , 50 Hz single phase power supply. Calculate the impedance, current through and power factor of the circuit.
Ans	Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$$X_L = \omega L = 2\pi fL = 2\pi \times 50 \times 0.3 = 94.25\Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}} = 31.83\Omega$$

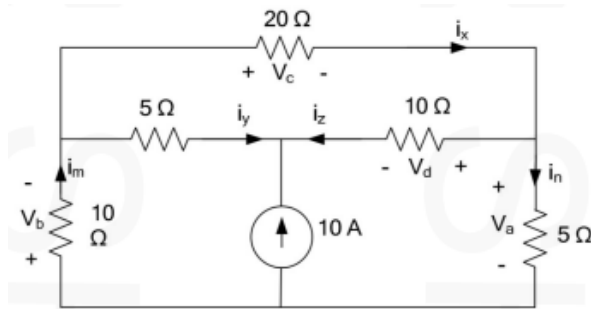
$$\text{Thus, Impedance } Z = \sqrt{10^2 + (94.25 - 31.83)^2} = 63.216\Omega$$

PART B

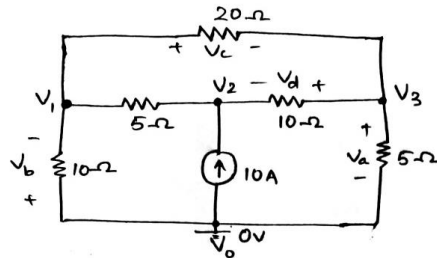
Answer any one full question from each module. Each question carries 14 marks

Module 1

Q6 Use nodal analysis to find voltages V_a , V_b , V_c , V_d



Ans



At node - 1

$$\frac{V_1}{10} + \frac{V_1 - V_2}{5} + \frac{V_1 - V_3}{20} = 0$$

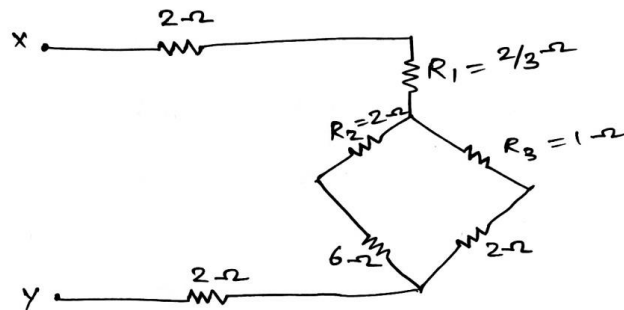
$$V_1 \left[\frac{1}{10} + \frac{1}{5} + \frac{1}{20} \right] - \frac{V_2}{5} - \frac{V_3}{20} = 0 \quad \text{--- ①}$$

At node - 2

$$\frac{V_2 - V_1}{5} + \frac{V_2 - V_3}{10} = 10$$

$$-\frac{V_1}{5} + V_2 \left[\frac{1}{5} + \frac{1}{10} \right] - \frac{V_3}{10} = 10 \quad \text{--- ②}$$

	At node - 3 $\frac{V_3}{5} + \frac{V_3 - V_2}{10} + \frac{V_3 - V_1}{20} = 0$ $-\frac{V_1}{20} - \frac{V_2}{10} + V_3 \left[\frac{1}{5} + \frac{1}{10} + \frac{1}{20} \right] = 0 \quad \text{--- (3)}$ Solving above 3 equations, we get $V_1 = 45.45 \text{ V} ; V_2 = 72.73 \text{ V} ; V_3 = 27.27 \text{ V}$ Thus, $\begin{aligned} V_a = V_3 &= 27.27 \text{ V} & V_c = V_1 - V_2 &= 18.18 \text{ V} \\ V_b = -V_1 &= -45.45 \text{ V} & V_d = V_3 - V_2 &= -45.46 \text{ V} \end{aligned}$
Q7	Find the equivalent resistance between the terminal X – Y in the network
Ans	

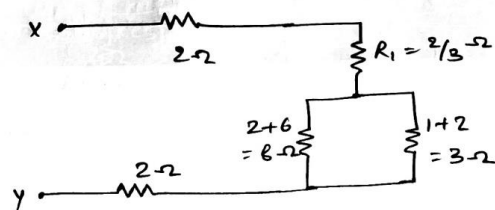


$$R_1 = \frac{4 \times 2}{4 + 2 + 6} = \frac{2}{3} \Omega$$

$$R_2 = \frac{4 \times 6}{4 + 2 + 6} = 2 \Omega$$

$$R_3 = \frac{2 \times 6}{4 + 2 + 6} = 1 \Omega$$

R_2 & 6Ω are in series; also R_3 & 2Ω are in series.



$\therefore$ Total Resistance across x and y

$$R_{xy} = 2 + \frac{2}{3} + (8 // 3) + 2$$

$$= \frac{14}{3} + \frac{8 \times 3}{8 + 3}$$

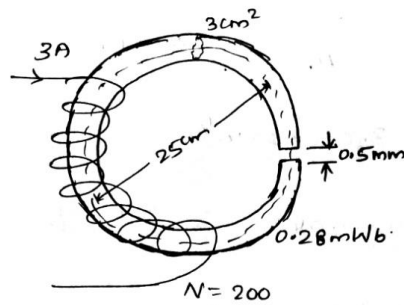
$$= \underline{\underline{6.85 \Omega}}$$

Module 2

- | | |
|----|---|
| Q8 | An iron ring has a cross sectional area of 3cm^2 and a mean diameter of 25cm . An air-gap of 0.5mm is cut across the section of the ring. The ring is wound with a coil of 200 turns |
|----|---|

through which a current of 3A. If the total magnetic flux is 0.28 mWb, find the relative permeability of the iron, assuming no magnetic leakage.

Ans



$$\phi = 0.28 \text{ mWb} = 0.28 \times 10^{-3} \text{ Wb}$$

$$N = 200 \text{ turns}$$

$$I = 3 \text{ A}$$

$$\begin{aligned} \text{We have, Reluctance, } S &= \frac{NI}{\phi} \\ &= \frac{200 \times 3}{0.28 \times 10^{-3}} \\ &= 2142857.143 \text{ AT/Wb} \end{aligned}$$

$$S = \frac{l}{\mu_0 \mu_r A}$$

$$\text{Here, } A = 3 \text{ cm}^2 = 3 \times 10^{-4} \text{ m}^2$$

Here there are 2 different materials.

$$\therefore S = \frac{l_a}{\mu_0 \mu_r A_a} + \frac{l_g}{\mu_0 A_g}$$

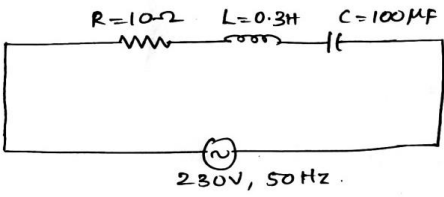
$$l_g = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$$

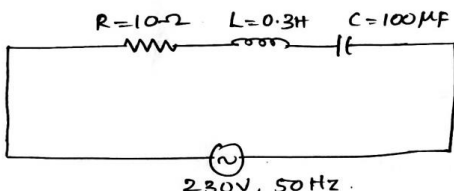
$$l_a = \pi d - l_g = 0.785 \text{ m}$$

$$\begin{aligned} 2142857.143 &= \frac{0.785}{4\pi \times 10^{-7} \times \mu_r \times 3 \times 10^{-4}} + \frac{0.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 3 \times 10^{-4}} \\ &= \frac{1}{4\pi \times 10^{-9} \times 3 \times 10^{-4}} \left[\frac{0.785}{\mu_r} + (0.5 \times 10^{-3}) \right] \end{aligned}$$

$$6.07638 \times 10^{-4} = \frac{0.785}{\mu_r} + (0.5 \times 10^{-3})$$

$$\frac{0.785}{\mu_r} = 3.07638 \times 10^{-4}$$

		$\therefore \text{Relative permeability of the material}$ $= \underline{\underline{2550.042}}$
Q9	a)	The instantaneous value of an alternating voltage is given by $v=110 \sin 314t$. Find the angular velocity, frequency, and time period of the voltage.
	b)	Differentiate between statically and dynamically induced emfs
Ans	a)	$v = 110 \sin 314t$ Comparing with general representation, $v = V_m \sin \omega t,$ Here Angular velocity, $\omega = \underline{\underline{314 \text{ rad/sec}}}$ frequency, $f = \frac{\omega}{2\pi}$ $= \underline{\underline{50 \text{ Hz}}}$ Time period, $T = \frac{1}{f}$ $= \underline{\underline{0.02 \text{ Sec}}}$
		$R=10\Omega$ $L=0.3H$ $C=100\mu F$  230V, 50Hz i) Impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2}$ $X_L = \omega L = 2\pi f L = 2\pi \times 50 \times 0.3 = \underline{\underline{94.25\Omega}}$ $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}} = \underline{\underline{31.83\Omega}}$ $Z = \sqrt{10^2 + (94.25 - 31.83)^2}$ $= \underline{\underline{63.216\Omega}}$

	b) EMF can be induced in 2 ways: 1) <u>Dynamically Induced EMF</u>: It is the EMF induced in a conductor or coil due to the relative motion between the conductor and the magnetic field. EMF will be induced dynamically when: ▪ A conductor is moved across a stationary field ▪ Field is moved across a stationary conductor ▪ Both the field and the conductor are moving 2) <u>Statically Induced EMF</u>: It is the EMF induced in a coil without any physical movement between the coil and the magnetic field. Statically induced EMFs are of 2 types: • <u>Self-induced emf</u>: It is the emf induced in a coil due to the change in current in the same coil itself is called self-induced emf. • <u>Mutually induced emf</u>: It is the emf induced in a coil due to the change in current in the neighbouring coil is called mutually induced emf.
Module 3	
Q10	A resistance of 10Ω, an inductance of 0.3 H and a capacitance of $100\mu\text{F}$ are connected in series across 230V, 50 Hz single phase supply. Calculate: i) the impedance of the circuit ii) Current through the circuit iii) Voltage across R, L and C and iv) Power consumed by the circuit
Ans	 i) Impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2}$ $X_L = \omega L = 2\pi f L = 2\pi \times 50 \times 0.3 = \underline{94.25\Omega}$ $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 100 \times 10^{-6}} = \underline{31.83\Omega}$ $Z = \sqrt{10^2 + (94.25 - 31.83)^2}$ $= \underline{\underline{63.216\Omega}}$

$$\text{ii) Current, } I = \frac{V}{Z}$$

$$= \frac{230}{63.216} = \underline{\underline{3.638 \text{ A}}}$$

$$\text{iii) Voltage across } R, = IR$$

$$= 3.638 \times 10$$

$$= \underline{\underline{36.38 \text{ V}}}$$

$$\text{Voltage across, } L, = IX_L$$

$$= 3.638 \times 94.25$$

$$= \underline{\underline{342.88 \text{ V}}}$$

$$\text{Voltage across } C, = IX_C$$

$$= 3.638 \times 31.83$$

$$= \underline{\underline{115.8 \text{ V}}}$$

$$\text{iv) Power consumed by the circuit} = VI \cos \phi$$

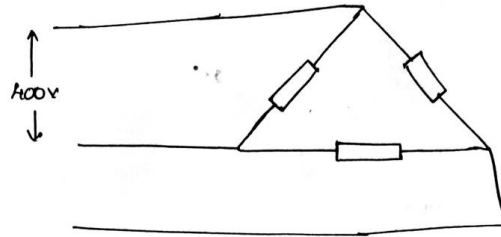
$$\cos \phi = \frac{R}{Z} = \frac{10}{63.216} = \underline{\underline{0.158 \text{ Lag}}}$$

$$\therefore P = 230 \times 3.638 \times 0.158$$

$$= \underline{\underline{132.362 \text{ W}}}$$

- Q11 A balanced delta connected 3 phase load is fed from a 3 phase, 50Hz power supply. The line current is 20A and the total power absorbed by the load is 10kW. Calculate (i) the impedance in each branch (ii) the power factor and) the total power consumed if the same impedances are star connected.

Ans



$$I_L = 20 \text{ A} , V_L = 400 \text{ V} , f = 50 \text{ Hz}$$

power absorbed, $P = 10 \text{ kW} = 10000 \text{ W}$

$$\sqrt{3} V_L I_L \cos \phi = 10000$$

$$\cos \phi = 0.72$$

i) Impedance in each branch (Z_{ph})

$$V_{ph} = V_L = 400 \text{ V}$$

$$I_{ph} = \frac{I_L}{\sqrt{3}} = 11.547 \text{ A}$$

$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = \underline{\underline{34.64 \Omega}}$$

ii) Power factor, $\cos \phi = \underline{\underline{0.72}}$

iii) Power consumed, if it is star connected,

$$\text{for } \Delta \text{ connection, } V_{ph} = \frac{V_L}{\sqrt{3}} = 230.94 \text{ V}$$

$$\cancel{I_{ph}} = \cancel{I_L} \quad I_{ph} = \frac{V_{ph}}{Z_{ph}} = 6.667 \text{ A}$$

$$\therefore \text{ power, } P = 3 V_{ph} I_{ph} \cos \phi = \underline{\underline{3.325 \text{ kW}}}$$